Project Abstract

Adaptive Continuous Authentication Using Behavioural Biometrics

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Abstract and Introduction

Abstract

The increasing reliance on web-based applications has exposed the limitations of conventional authentication mechanisms such as passwords, one-time passwords, and static multi-factor authentication. These techniques verify user identity only at the time of login and remain vulnerable to threats including credential theft, phishing attacks, and session hijacking. To address these challenges, this project proposes an adaptive continuous authentication system based on behavioural biometrics for web environments.

The system continuously monitors user interaction behaviour, including keystroke dynamics and mouse movement patterns, during active sessions. Behavioural data is captured in real time through client-side scripting and securely transmitted to a backend server for storage and analysis. The collected data forms the foundation for constructing individual behavioural profiles and enabling future integration of machine learning and deep learning techniques for anomaly detection, behaviour drift handling, and adaptive authentication decisions. An alert mechanism is also planned to notify legitimate users via email when suspicious behaviour is detected. The proposed approach enhances security while maintaining a seamless user experience.

Introduction

Authentication plays a pivotal role in safeguarding digital systems against unauthorised access. Traditional authentication methods, including passwords, PINs, one-time passwords, and multi-factor authentication, perform identity verification solely during the login process. After initial verification, the system assumes that the authenticated user remains active, which introduces potential vulnerabilities. This assumption makes systems susceptible to session hijacking, insider threats, unattended terminals, and other security breaches that occur post-login.

Behavioural biometrics presents a promising solution for continuous and non-intrusive authentication by analysing *how* users interact with a system rather than relying solely on what credentials they provide. Patterns such as typing speed, key hold duration, typing rhythm, mouse movement trajectories, and click patterns are inherently unique and difficult for attackers to replicate, making them ideal for continuous verification.

This project focuses on developing a **web-based adaptive continuous authentication system** that actively monitors user behaviour, collects and stores interaction data, and lays the foundation for intelligent authentication mechanisms. The system aims to detect behavioural anomalies, handle behaviour drift over time, and adapt authentication decisions dynamically to enhance security without compromising user convenience.

Problem Statement and Objectives

Problem Statement

Conventional authentication systems are static and limited to the initial login, failing to verify user identity continuously during active sessions. If an attacker gains access to an already authenticated session, traditional systems are incapable of detecting abnormal behaviour or preventing misuse. Existing continuous authentication solutions either lack adaptability to changes in user behaviour over time or are intrusive, negatively affecting the user experience. Therefore, there is a need for a **web-based, adaptive, and user-friendly continuous authentication system** that evaluates user identity continuously using behavioural biometrics while remaining unobtrusive.

Objectives of the Project

The primary objectives of this project are:

- To design and implement a web-based continuous authentication system using behavioural biometrics.
- To capture user interaction data, including keystroke dynamics and mouse movement patterns, in real time.
- To store behavioural data securely in a structured format for further analysis.
- To provide a foundation for adaptive authentication through machine learning and deep learning techniques.
- To support behaviour drift detection and enable dynamic authentication decisions.
- To plan and implement a notification mechanism for alerting users about suspicious behaviour.
- To ensure the system remains non-intrusive while providing robust security for web applications.

Existing System – Pros and Cons

Existing System

Existing authentication systems primarily rely on passwords, PINs, OTPs, and multi-factor authentication mechanisms that verify identity only during login.

Advantages

- Simple and widely adopted
- Easy to implement and manage
- Low computational overhead

Disadvantages

- No continuous verification after login
- Vulnerable to credential theft and session hijacking
- Password reuse and phishing attacks are common
- Unable to adapt to changes in user behaviour
- No mechanism to detect unauthorised access during active sessions

Proposed System – Pros

Proposed System

The proposed system introduces a **web-based adaptive continuous authentication framework** that continuously verifies user identity using behavioural biometrics. The system is built on a client–server architecture with a Flask backend for processing and storage, and a frontend interface for capturing keystroke and mouse interaction data.

- The framework is designed to:
- Collect real-time behavioural data through JavaScript event listeners.
- Store interaction data in a structured JSON format for analysis.
- Construct individual user behavioural profiles.
- Enable future integration with machine learning and deep learning models for anomaly detection and drift analysis.
- Notify users of suspicious activity via planned alert mechanisms.

Advantages of the Proposed System

- Continuous and real-time verification of user identity throughout active sessions
- Non-intrusive authentication without disrupting user workflow
- Enhanced security against session hijacking and credential misuse
- Ability to adapt to behavioural changes and detect behaviour drift
- Supports advanced analytics using machine learning and deep learning
- Enables alert-based notifications for detected anomalies

Software and Frameworks Needed

Software Requirements

- Operating System: Windows 10/11
- Programming Languages: Python 3.14, JavaScript

Frontend Technologies

- HTML5
- CSS3
- JavaScript (ES6+)

Backend Framework

- Flask (Python web framework)
- Flask-CORS for cross-origin request handling

Data Storage

- JSON format for behavioural interaction logs
- Local file system for initial storage; can be upgraded to database storage in the future

Planned Tools for Future Integration

- Machine learning libraries: scikit-learn, XGBoost
- Deep learning frameworks: TensorFlow or PyTorch
- Email services for alert notifications (SMTP/third-party APIs)

References

Relevant Articles and Research Papers

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3. Wyciślik, Ł., Wylężek, P., & Momot, A. (2024). The Improved Biometric Identification of Keystroke Dynamics Based on Deep Learning Approaches. *Sensors*, *24*(12), 3763. https://doi.org/10.3390/s24123763

Frameworks and Tools for Custom Development

- Flask framework for backend development
- JavaScript event listeners for behavioural data capture
- RESTful APIs for client–server communication

Dataset

A custom dataset is generated by capturing user behavioural interaction data, including keystroke dynamics, key hold durations, typing rhythm, and mouse movement patterns. The data is stored in structured JSON format and will be utilised for behavioural modelling, drift detection, and adaptive authentication analysis in later stages of the project.